Uncertainty-Guided Agentic Reasoning: Investigating the Impact of Uncertainty Scores on LLM Agents

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***Abstract*—Current agentic systems demonstrate outstanding performance among various types of tasks, including code generation and validation. However due to probabilistic nature of models, controversial information, and inconsistency in dateset quality, Large Language Models tend to hallucinate, generating structurally correct, but logically false code snippets. Hallucination detections methods vary in their methodology, however one specific subset stands out among others. Present papers demonstrated a property of hidden states to store information related to models awareness. Using this insights it is possible to detect, but not affect generation of hallucinations. In this paper we plan to investigate performance of hallucination aware agentic loops. Resulting system will not only be able to detect hallucinated samples, but also apply this knowledge for generation of correct code**

***Index Terms*—LLM agents, uncertainty-guided reasoning, uncertainty scores, hallucination detection, hidden states**

I. Introduction

Current Large Language Models (LLMs) have demonstrated outstanding performance in various types of tasks, including question answering [1], summarization [2], and most importantly software engineering [3]. Integration of LLMs in agentic loops allowed to fully automate many processes and substantially increase efficiency and quality of software engineers work.

Even so, truthfulness and quality of LLMs generations still suffers from hallucinations problem. Hallucination refers to situation when model answers are structurally correct, but logically or factuality false. This may happen due to many hard to manage factors [4] such as poor quality of training data, problems with context size or even corrupted context by user itself. Hallucinated model output often leads to wrong conclusions, and misleading instructions, that may confuse user. Agentic systems suffer from this even more - one poor design decision or incorrect tool use may corrupt the whole context, heavily influencing future actions, and in most cases leading to poor software quality or completely broken project.

Present papers manly focussed on hallucination detection [5], [6], [7], [8] in various ways of estimating model uncertainty. One of proposed approaches uses model output token probabilities to quantify uncertainty, hence possibly detection hallucinated sample. Another approaches use method of sampling different responses on same input promt, focusing on estimating uncertainty of specific fact, rather that some token. One of the most promising ways of uncertainty estimation is to use internal model representations, collected from various layers, hence capturing hallucination, even before token generated. Many approaches also explore combinations of proposed methods to train a separate simple MLP or Attention Head, that classify false and true samples.

However, even with sufficient number of papers exploring hallucination detection, there are still limited number of research actually fixing false generations [9], [10]. Even so, in most cases this solutions only oriented on simple Q/A tasks or one promt interaction with LLMs, but not complex long context interactions with models, that the case for coding agents.

This research aims to investigates the role of uncertainty score on agent behaviour by integrating various ways of hallucination estimators directly into agentic loops. As a result we present a complete agentic system, capable of detection and fixing hallucinated samples, hence increasing overall performance.

Main contribution of this paper is the following:

- **Hallucination Aware Agentic Loop** - we propose hallucination aware agentic loop, that uses various uncertainty scores, for detecting and fixing hallucinated samples.

II. Literature Review

A. Hallucination in Coding Tasks

Hallucinations have various definitions. One of them was proposed by [11] together with **CodeHalu** benchmark. Authors define hallucination as ”syntactically correct and even semantically plausible, but may not execute as expected or fulfill specified requirements”.

Systematic review of hallucination in software engineering tasks by [12] presents complete taxonomy on various hallucination types in code generation tasks. Additionally authors presented **HalluCode** benchmark, containing 5563 Python coding tasks with reference and hallucinated solutions. Similarly to [12] in [13] authors provide survey on hallucination detection and mitigation across various types of models. Authors also provide taxonomy on hallucination dividing them on four main categories: **Contextual Disconnection**, **Semantic Distortion**,

Content Hallucination, and **Factual Inaccuracy** Another review [4] categorizes hallucinations on two types, factuality hallucination and faithfulness hallucinations. In addition paper review mitigation strategies, including **data filtering, RAG, and targeted optimizations**

Summing all up we can conclude that hallucination refers to situation, then **models output have correct code structure but fails to solve specified problem in efficient way, due to misalignment in internal and external knowledge and poor context quality**

B. Uncertainty Estimations

One of the ways to detect hallucinations was proposed by [8]. Authors presented a novel fact checking pipeline **Claim Conditioned Probability** designed especially for checking of atomic LLM claims.

Another way to detect hallucination firstly was introduced in [5]. SPALMA uses internal model representations to estimate models uncertainty score. THis paper demonstrated ability of the models to store uncertainty information inside. In other paper [14] authors explored approach proposed by [5]. Findings show that the most insights for uncertainty estimations were acquired from models **middle layers** (10-20 for LLama models).

Such approach demonstrated high performance, hence became vary popular among other researchers like [15] that presented HaloScope that also uses features extracted from internal representations, that later passed to simple classifier Similarly, [7] trained special **Uncertainty Quantification Heads** on hidden states of model to detect hallucinated samples.

In [16] authors proposed **Lookback Lense**, a light weight method, that uses attention weights for input and generated tokens. Proposed approach matches performance with more complicated detectors, while using simple linear classifier. Authors of [6] made one step further by analyzing not just internal representations but their change among layers, proposing **ICR Probe** and proving a way to distinguish hallucination cause based on **ICR Score** value

Interesting methodology was proposed in [17]. Authors assumes that dispersion or variance of internal representations across layers may reflect internal model uncertainty. Proposed approach avoids large resource consumption and shows effective generalization across different domains

C. Uncertainty Estimations in Coding

In [18] authors propose to use uncertainty estimation as a proxy value for checking LLMs generated code. In paper authors adapted NLP methods for software engineering tasks and added a check semantic equivalence check for ensuring that model understood the task. In contrast [19] showed that natural-language semantic entropy does not transfer well to code, instead proposing **functional equivalence methods**, that include coding analog of semantic entropy. Another approach [9] uses combination of features to train small transformer model for line-level hallucination detection. It outperforms existing UQ baselines across several code domains.

D. Uncertainty Estimations in Agents

Only a few paper review role of uncertainty estimates on agentic work. One of such paper [10] proposes **SAUP**, framework that propagates uncertainty through each step of agents reasoning, achieving state-of-art performance. UALA by [20] tried to integrate uncertainty estimate directly into REACT agent. Generated response evaluated on model uncertainty, and based on result continues with CoT or uses external tool for accessing information

E. Research Gap

Reviewed papers demonstrated that it is possible to detect model hallucinations in various ways, however current mitigation techniques manually work with simple one-shot interactions with users. Only limited number of papers explore influence of uncertainty estimators on coding agents. Additionally, most of approaches strictly tied to specific architecture, hence for each new model fine-tuning of classifiers and features from scratch required. This thesis ties to address this gap by exploring effect of various uncertainty scores on agent behaviour in software engineering tasks.

III. Methodology

IV. Experiments

V. Discussion

VI. Conclusion

VII. Acknowledgment

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